

AYUSH KUMAR RAY

CSE Student | AI/ML Engineer | Data Analyst | Multi-Agent Systems | Full-Stack Developer

Parul University, Vadodara, Gujarat

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Professional Summary

CSE student (2023–2027) with hands-on expertise across AI/ML engineering, data analytics, and full-stack development. Built LLM-powered multi-agent systems (LangChain, LangGraph), end-to-end ML pipelines (**93% F1-score** on 284K transactions), and data analysis workflows with Python, SQL, and Power BI. Solved **1500+ DSA problems**. Holds **5 certifications**: AWS Cloud, IBM AI, ServiceNow CSA & CAD, Smart Coder Bronze.

Education

Parul University

Bachelor of Technology in Computer Science Engineering

2023 – 2027

Vadodara, Gujarat

- CGPA: 7.84 / 10 Relevant Coursework: DSA, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering

Technical Skills

Languages: Python, Java, C++, JavaScript, SQL, Dart

AI / ML Engineering: Scikit-learn, SMOTE, Feature Engineering, Model Training & Evaluation, Inference Pipelines, Hyperparameter Tuning, NLP, Whisper (STT), Pandas, NumPy

Agentic AI & LLM: LangChain, LangGraph, Multi-Agent Systems, LLM Integration (OpenAI / Anthropic APIs), RAG, FAISS Vector Store, Prompt Engineering, LLM Function Calling

Data Analysis: Pandas, NumPy, SQL Query Optimization, Exploratory Data Analysis (EDA), Statistical Analysis, Data Cleaning, Data Wrangling

Data Visualization: Power BI, Matplotlib, Seaborn, Dashboard Design

Backend & APIs: Django REST Framework, FastAPI, Spring Boot, RESTful APIs, WebSockets, JWT Auth, Microservices

Cloud & DevOps: AWS (EC2, S3, RDS), Firebase, Linux, Git, GitHub, CI/CD

Databases: PostgreSQL (PostGIS), MySQL, MongoDB, Database Design

Problem Solving: Data Structures & Algorithms, Dynamic Programming, Graph Algorithms, Complexity Analysis, OOP

Projects

AI-Powered Emergency Response & Hospital Coordination System | *Python, Django, LangChain, LangGraph, PostgreSQL, AWS2023*

- Designed a **multi-agent system** (LangChain + LangGraph) with a Triage Agent and Dispatch Agent coordinating in real time – Triage Agent uses **LLM function calling + Whisper STT** to classify symptom severity into 3 tiers and autonomously trigger dispatch.
- Built a **PostGIS** geospatial allocation engine performing EDA-style scoring on live hospital data (distance, bed availability, ICU match); assigns optimal hospital in under **1.5 seconds**.
- Developed RESTful backend (Django REST Framework + WebSockets) handling 4 actor types; avg API response under **200ms**. Deployed on **AWS EC2/RDS** with Firebase push notifications.

Credit Card Fraud Detection – End-to-End ML & Data Analysis Pipeline | *Python, Scikit-learn, Pandas, NumPy, SMOTE, SQL, 2023*

- Performed **EDA and data wrangling** on **284K+ transactions** (fraud rate: 0.17%): statistical analysis, data cleaning, correlation analysis, and feature engineering to surface fraud signals.
- Applied **SMOTE** to rebalance classes and trained 3 classifiers (Random Forest, Logistic Regression, Gradient Boosting) with Grid Search hyperparameter tuning; best model: **93% F1-score, 0.97 ROC-AUC**.
- Built a **Power BI dashboard** visualizing fraud trends, transaction distributions, and model performance metrics (Precision, Recall, Confusion Matrix) for stakeholder reporting.

RAG-Powered Document Q&A Chatbot | *Python, LangChain, FAISS, OpenAI API, FastAPI, React*

2025

- Built a **Retrieval-Augmented Generation (RAG)** pipeline ingesting PDFs into a **FAISS vector store** with OpenAI embeddings; source-cited query responses under **2 seconds** on 50-page documents.
- Exposed via **FastAPI** REST backend + React frontend with **prompt engineering** for response accuracy; supports multi-document upload and persistent chat history.

Competitive Programming

Platforms: LeetCode, CodeChef, HackerRank, Smart Interviews

Problem Solving: Solved **1500+ problems** – Dynamic Programming, Graph Algorithms, Trees, Recursion, Greedy, Sliding Window.

Achievement: **Smart Coder (Bronze)** certified by Smart Interviews for DSA proficiency.

Certifications

AWS Academy Graduate – Cloud Foundations
Artificial Intelligence Fundamentals
ServiceNow Certified System Administrator (CSA)
ServiceNow Certified Application Developer (CAD)
Smart Coder (Bronze)

Amazon Web Services
IBM SkillsBuild
ServiceNow
ServiceNow
Smart Interviews